

Safety & Vertical Curves

Crest Vertical Curves

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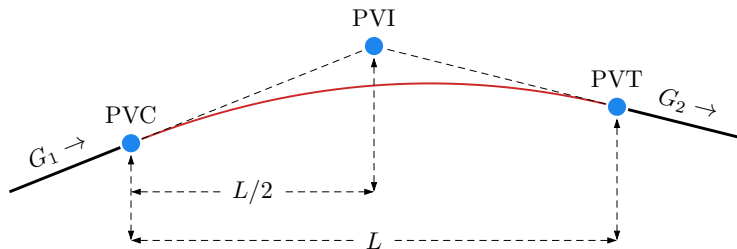
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Vertical Alignment in Highway Design

- Vertical alignment refers to the elevation of a roadway along its length
- Involves the design of:
 - Grades (slopes)
 - Vertical curves connecting grades

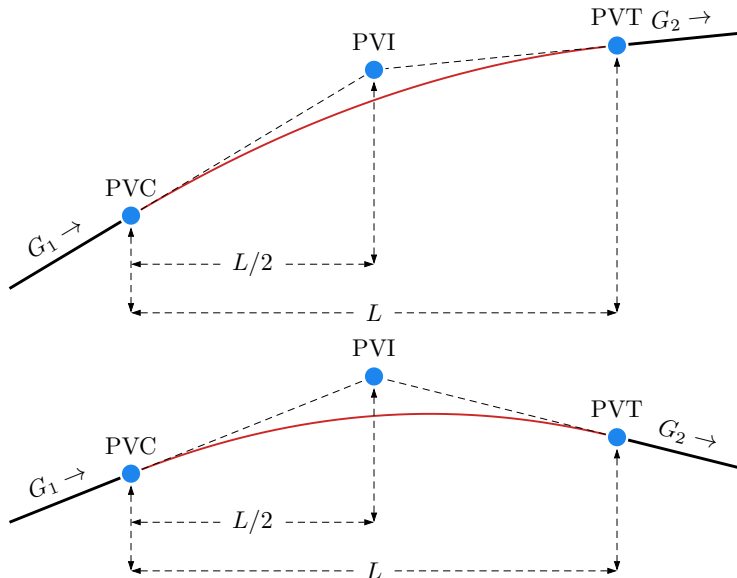
Definitions for Crest Vertical Curves

- PVC **Point of Vertical Curvature** (start)
- PVI **Point of Vertical Intersection**
- PVT **Point of Vertical Tangent** (end)
- G_1 initial grade
- G_2 final grade
- L curve length (m)
- L is measured *horizontally* (not along the curve)



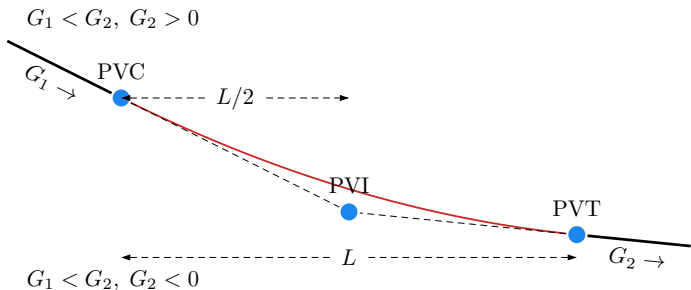
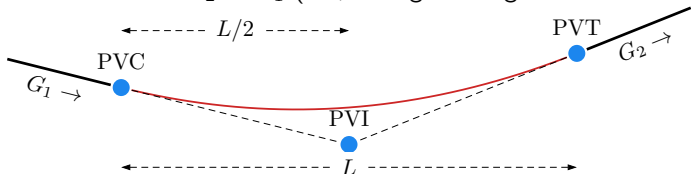
Crest Vertical Curves

A **crest** vertical curve has $G_2 < G_1$ (i.e., final grade is less than initial grade)



Sag Vertical Curves

A **sag** vertical curve has $G_2 > G_1$ (i.e., final grade is greater than initial grade)



Vertical Curve Equation

- Vertical curves are designed as **pieces of parabolas** (like horizontal curves are pieces of circles)
- The elevation at any point is given by:

$$y = y_{\text{PVC}} + G_1 \cdot x + \frac{G_2 - G_1}{2L} \cdot x^2$$

where:

- y is the elevation at distance x from the PVC
 - y_{PVC} is the elevation at the PVC
 - x is the distance from the PVC ($0 \leq x \leq L$)
 - L is the length of the vertical curve
- If it's a crest curve, for what value of x does the curve reach its highest point? (Hint: use the first-order condition)

Properties of Crest Vertical Curves

- Grade at any point x from PVC:

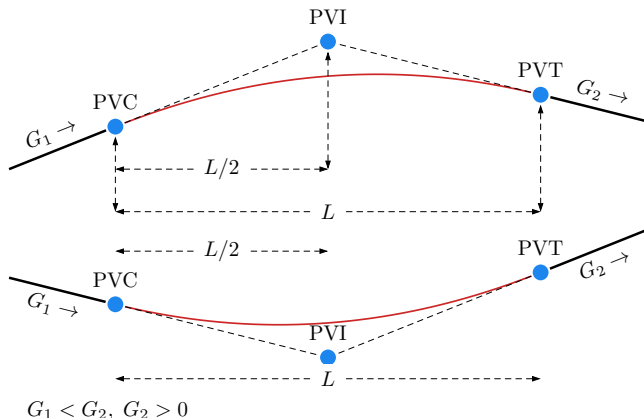
$$\frac{\partial y}{\partial x} = G_1 + \frac{G_2 - G_1}{L} \cdot x_{\max} = 0$$
$$x_{\max} = \frac{G_1 \cdot L}{G_1 - G_2}$$

- If $G_2 < 0$, then $x_{\max} < L$ and the curve has a maximum point.
- If $G_2 > 0$, then $x_{\max} > L$ and the curve has no maximum point.

Safety Considerations

Safety Considerations for Vertical Curves

- Sag and crest vertical curves have different safety considerations.
- What are they?
- We will focus on **crest** vertical curves.

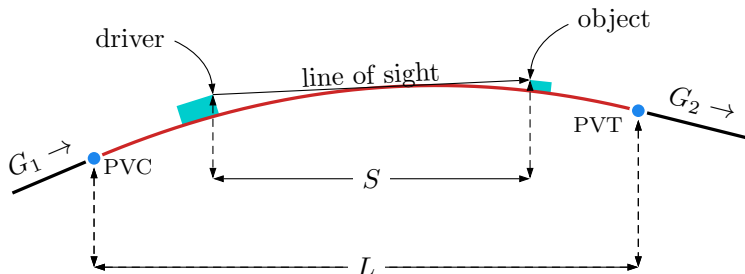


Sight distance on crest vertical curves

On crest vertical curves, it is important to be able to see over the curve. Let

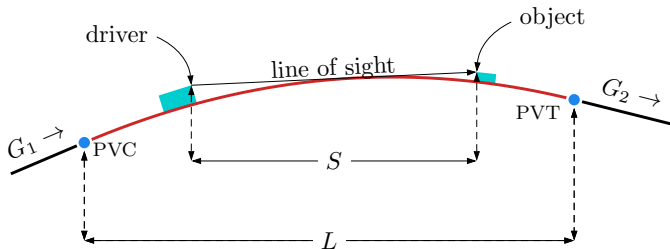
- S be the sight distance: *how far over the crest the driver can see some object of a certain height.*
- h_1 be the assumed height of the driver's eyes
- h_2 be the assumed height of the object
- $A = |G_1 - G_2|$ be the absolute change in grade
- $z = 2(\sqrt{h_1} + \sqrt{h_2})^2 / A$ is a parameter.

If $z < L$, then $S = \sqrt{zL}$.

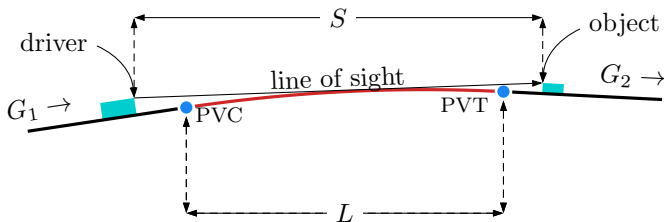


Sight distance on crest vertical curves

- $z = 2(\sqrt{h_1} + \sqrt{h_2})^2 / A$
- If $z < L$, then $S = \sqrt{zL}$.



- If $z > L$, then $S = (z + L)/2$.



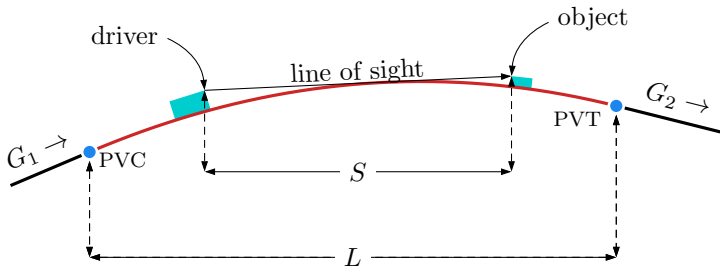
Minimum Curve Length Requirements: Case 1

$(x_{SSD} < L_{min})$

- You want to ensure $S \geq x_{SSD}$.
- Case 1: Check if $x_{SSD} > z$.
- Minimum length to provide adequate sight distance:

$$L_{min} = \frac{x_{SSD}^2}{z}$$

- This means the driver is on the curve when they see the hazard.



- In practice, most designs use Case 1.

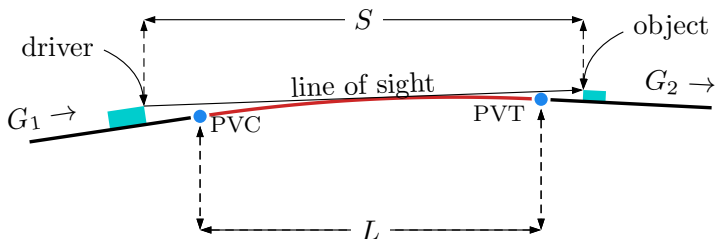
Minimum Curve Length Requirements: Case 2

$(x_{SSD} > L_{min})$

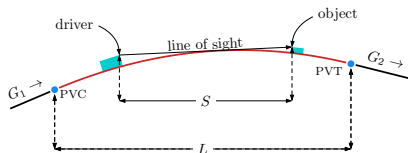
- Case 2: Check if $x_{SSD} < z$.
- Minimum length to provide adequate sight distance:

$$L_{min} = 2x_{SSD} - z$$

- This means the driver is off the curve when they see the hazard.



Practice Problem - Crest Vertical Curve



- A crest vertical curve has:
 - $G_1 = +0.04$
 - $G_2 = -0.02$
 - desired sight distance $x_{SSD} = 120$ m
 - $h_1 = 1.08$ m
 - $h_2 = 0.6$ m
- Find:
 - $A = |G_1 - G_2|$
 - $z = 2(\sqrt{h_1} + \sqrt{h_2})^2 / A$
 - the minimum curve length $L_{\min}$
- Which minimum-length case applies?

Solution - Crest Vertical Curve

- First compute the grade difference:

$$\begin{aligned}A &= |G_1 - G_2| \\&= |0.04 - (-0.02)| \\&= 0.06\end{aligned}$$

- Next compute z :

$$\begin{aligned}z &= \frac{2(\sqrt{h_1} + \sqrt{h_2})^2}{A} \\&= \frac{2(\sqrt{1.08} + \sqrt{0.6})^2}{0.06} \\&\approx 109.7 \text{ m}\end{aligned}$$

- Since $x_{SSD} = 120 \text{ m} > z = 109.7 \text{ m}$, use the **Case 1** formula:

$$L_{\min} = \frac{x_{SSD}^2}{z}$$

- Therefore,

$$\begin{aligned}L_{\min} &= \frac{120^2}{109.7} \\&\approx 131.3 \text{ m}\end{aligned}$$

- So the minimum required curve length is about **131 m**